

1. Consider:

```
x = torch.tensor([2.0], requires_grad=True)
y = x**2 + 3*x + 1
y.backward()
```

What is the value inside x.grad?

Explain how PyTorch computed it step-by-step.

Answer : The value of x.grad is 7.0. After running this code and selecting requires_grad=True enables auto gradient tracking by creating a computational graph. By applying the chain rule the autograd computes each node separately . like $x=2 \rightarrow a=x^2 \rightarrow b=3x \rightarrow c=a+b \rightarrow d=c+1 \rightarrow y$, so the whole thing becomes 7.0 when called backward() from y.

2. What happens in this case?

```
x = torch.tensor([2.0])
y = x**2

y.backward()
```

Why does it fail?

Answer : In this case a Runtime error occurs . We didn't enable gradient tracking for x here . For this no computational graph will form. By calling the backward() function the chain derivation will not be possible to obtain in absence of the computational graph, this is why this code will fail.